

Flow Rate Velocity Controller

Consider the controller output at sample k to be $u(k)$. In velocity-form PID control, the current output is updated from the previous output by adding an incremental change $\Delta u(k)$:

$$u(k) = u(k-1) + \Delta u(k)$$

PID equation

$$u(k) = K_p \cdot e(k) + K_i \cdot T_s \cdot \sum_{i=0}^k e(i) + K_d \cdot \frac{e(k) - e(k-1)}{T_s}$$

Where $e(k)$ is the error for a give sample and T_s is the sample time.

$$\Delta u(k) = u(k) - u(k-1)$$

$$\Delta u(k) = K_p \cdot (e(k) - e(k-1)) + K_i \cdot T_s \cdot \left(\sum_{i=0}^k e(i) - \sum_{i=0}^{k-1} e(i) \right) + K_d \cdot \left(\frac{e(k) - 2e(k-1) + e(k-2)}{T_s} \right)$$

$$\Delta u(k) = K_p \cdot (e(k) - e(k-1)) + K_i \cdot T_s \cdot e(k) + K_d \cdot \left(\frac{e(k) - 2e(k-1) + e(k-2)}{T_s} \right)$$

PI velocity controller is adopted as the flow rate controller, as it provides a suitable balance between implementation simplicity, control responsiveness, and steady-state accuracy, while avoiding the increased noise sensitivity associated with derivative action.

PI velocity equation

$$u(k) = u(k-1) + K_p \cdot (e(k) - e(k-1)) + K_i \cdot T_s \cdot e(k)$$

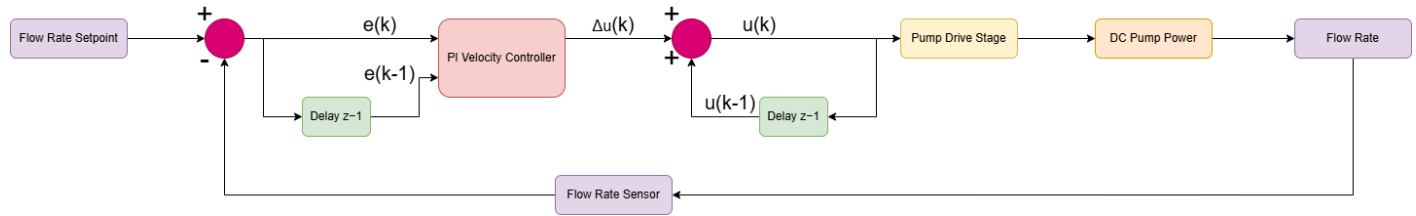


Figure 1: Control block diagram